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LAB ASSIGNMENT – 3

Objectives :

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

DDL Questions (Data Definition Language)

1.) Create the TEACHER table with the following attributes:
(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```sql
CREATE DATABASE HARSH_LABASSIGN_3;
USE HARSH_LABASSIGN_3;
CREATE TABLE TEACHER(
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

2.) Create the SALES table with the following attributes:
(Sales_No, Salesname, Branch, Salesamount, DOB).

```sql
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount DECIMAL(10,2),
    DOB DATE
);
```

3.) Create the DEPARTMENTS table with the following attributes: (dept_no, dept_name, dept_location).

```sql
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

4.) Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```sql
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

DML Questions (Data Manipulation Language)

From TEACHER Table:

1.) Insert five records into the TEACHER table.

```sql
INSERT INTO TEACHER VALUES
('Harsh',1,'2018-06-15','Mathematics','Delhi',50000),
('Harsh',2,'2019-07-20','Commerce','Mumbai',45000),
('Nirvikalp',1,'2017-03-12','Mathematics','Delhi',55000),
('Rohan',3,'2020-01-10','Physics','Chennai',40000),
('Aaryan',2,'2016-09-05','Commerce','Mumbai',48000);

SELECT * FROM TEACHER;
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Krish	1	2018-06-15	Mathematics	Delhi	50000.00
Harsh	2	2019-07-20	Commerce	Mumbai	45000.00
Nirvikalp	1	2017-03-12	Mathematics	Delhi	55000.00
Rohan	3	2020-01-10	Physics	Chennai	40000.00
Aaryan	2	2016-09-05	Commerce	Mumbai	48000.00

2.) Give an increment of 25% salary to teachers working in the Mathematics department.

```
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
SELECT* FROM TEACHER;
```

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Krish	1	2018-06-15	Mathematics	Delhi	62500.00
Harsh	2	2019-07-20	Commerce	Mumbai	45000.00
Nirvikalp	1	2017-03-12	Mathematics	Delhi	68750.00
Rohan	3	2020-01-10	Physics	Chennai	40000.00
Aaryan	2	2016-09-05	Commerce	Mumbai	48000.00

3.) Give an increment of 15% salary to teachers working in the Commerce department.

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SELECT* FROM TEACHER;
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Krish	1	2018-06-15	Mathematics	Delhi	62500.00
Harsh	2	2019-07-20	Commerce	Mumbai	51750.00
Nirvikalp	1	2017-03-12	Mathematics	Delhi	68750.00
Rohan	3	2020-01-10	Physics	Chennai	40000.00
Aaryan	2	2016-09-05	Commerce	Mumbai	55200.00

From SALES Table

1.) Insert five records into the SALES table.

```
INSERT INTO SALES VALUES
(101,'Krish','North',20000,'1998-12-21'),
(102,'Mukul','South',30000,'1997-05-10'),
(103,'Keshav','East',25000,'1996-12-11'),
(104,'Vaibhav','West',40000,'1995-08-15'),
(105,'Manav','North',15000,'1999-12-05');
SELECT * FROM SALES;
```

OUTPUT :

Sales_No	Salesname	Branch	Salesamount	DOB
101	Krish	North	20000.00	1998-12-21
102	Mukul	South	30000.00	1997-05-10
103	Keshav	East	25000.00	1996-12-11
104	Vaibhav	West	40000.00	1995-08-15
105	Manav	North	15000.00	1999-12-05
NULL	NULL	NULL	NULL	NULL

2.) Calculate the total sales amount in each branch.

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT :

Result Grid | Filter

Branch	Total_Sales
North	35000.00
South	30000.00
East	25000.00
West	40000.00

3.) Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Avg_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT :

Result Grid | Filter Ro

Branch	Avg_Sales
North	17500.000000
South	30000.000000
East	25000.000000
West	40000.000000

4.) Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

```
SELECT Salesname,
DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT:

Salesname	DOB
Krish	21-Dec-98
Keshav	11-Dec-96
Manav	05-Dec-99

5.) Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

OUTPUT:

Salesname	DOB
Vaibhav	1995-08-15
Krish	1998-12-21
Keshav	1996-12-11
Manav	1999-12-05
Mukul	1997-05-10

From JOIN Operations:

First We Insert Records for JOIN:

Insert into DEPARTMENTS

```sql
INSERT INTO DEPARTMENTS VALUES
(1, 'Mathematics', 'Delhi'),
(2, 'Commerce', 'Mumbai'),
(3, 'Physics', 'Gurugram');
SELECT* FROM DEPARTMENTS;
```

OUTPUT :

Result Grid | Filter Rows:

dept_no	dept_name	dept_location
1	Mathematics	Delhi
2	Commerce	Mumbai
3	Physics	Gurugram
NULL	NULL	NULL

Insert into EMPLOYEES

```sql
INSERT INTO EMPLOYEES VALUES
(1,'Ankit',50000,1),
(2,'Bhavesh',45000,2),
(3,'Krish',60000,1),
(4,'Dipanshu',35000,3),
(5,'Ashu',70000,2);
SELECT * FROM EMPLOYEES;
```

emp_id	emp_name	emp_salary	dept_no
1	Ankit	50000.00	1
2	Bhavesh	45000.00	2
3	Krish	60000.00	1
4	Dipanshu	35000.00	3
5	Ashu	70000.00	2
NULL	NULL	NULL	NULL

1.) Display employee details and department details where the department numbers are the same in both tables.

```sql
SELECT *
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

	emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
▶	1	Ankit	50000.00	1	1	Mathematics	Delhi
	3	Krish	60000.00	1	1	Mathematics	Delhi
	2	Bhavesh	45000.00	2	2	Commerce	Mumbai
	5	Ashu	70000.00	2	2	Commerce	Mumbai
	4	Dipanshu	35000.00	3	3	Physics	Gurugram

2.) Display the employee name and department name using a LEFT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

	emp_name	dept_name
▶	Ankit	Mathematics
	Bhavesh	Commerce
	Krish	Mathematics
	Dipanshu	Physics
	Ashu	Commerce

3.) Display the employee name and department name using a RIGHT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT :

	emp_name	dept_name
▶	Ankit	Mathematics
	Krish	Mathematics
	Bhavesh	Commerce
	Ashu	Commerce
	Dipanshu	Physics

4.) Display the details of employees who draw a salary greater than the average salary.

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

OUTPUT :

emp_id	emp_name	emp_salary	dept_no
3	Krish	60000.00	1
5	Ashu	70000.00	2
NULL	NULL	NULL	NULL

DCL Questions (Data Control Language):

1.) Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

FIRST WE HAVE TO CREATE A USER

```
CREATE USER 'departments_user'@'localhost'
IDENTIFIED BY '1234';
```

THEN WE GRANT ALL PRIVILEGES ON THE EMPLOYEE TABLE TODEPARTMENTS

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Krish	1	2018-06-15	Mathematics	Delhi	50000.00
Harsh	2	2019-07-20	Commerce	Mumbai	45000.00
Nirvikalp	1	2017-03-12	Mathematics	Delhi	55000.00
Rohan	3	2020-01-10	Physics	Chennai	40000.00
Aaryan	2	2016-09-05	Commerce	Mumbai	48000.00

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...
GRANT ALL PRIVILEGES ON `krish_labassign_3...

2.) Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON KRISH_LABASSIGN_3.EMPLOYEES
TO 'departments_user'@'localhost';
```

OUTPUT:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT ALL PRIVILEGES ON `krish_laba GRANT AL

3.) Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON KRISH_LABASSIGN_3.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...

4.) Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

First, we have to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON KRISH_LABASSIGN_3.EMPLOYEES
TO 'departments_user'@'localhost';
```

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT ALL PRIVILEGES ON `krish_laba GRANT AL

10 Now, we revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT
ON KRISH_LABASSIGN_3.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT :

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...